Please print, cut, and paste the procedures into your practical notebook 2 before the scheduled date on Monday, November 2, 2026.

Please make sure to cut and paste them neatly. Thank you!

Lesson 3: The States of Matter **1**

DETECTION OF GASES

A. Testing Procedure for Hydrogen Gas

Principle

Hydrogen gas is highly flammable and, when mixed with oxygen (air), it ignites rapidly upon exposure to a flame, producing a distinctive sound.

Safety Precautions

1. Keep the gas away from any open flames
2. Wear safety googles at all times.

Apparatus and Materials

- Test tube containing the gas sample to be tested
- Lighted wooden splint
- Matches or a lighter

Procedure

1. Keeping your thumb over the mouth, bring a lighted splint near the opening.
2. Quickly remove your thumb as the gas hits the flame.

   Expected Observation: It will make a loud "squeaky pop" sound.

Observation and Conclusion **2**

Observation:

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Conclusion:

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B. Testing Procedure for Carbon Dioxide Gas

Principle

Carbon dioxide is **colorless** and **odorless gas**. It does not support combustion (burning). The characteristic test involves bubbling the gas through **limewater (**calcium hydroxide) which reacts to form calcium carbonate.

Calcium hydroxide + carbon dioxide → Calcium Carbonate + water

Safety Precautions

- Handle the limewater carefully as it is an irritant.

Apparatus and Materials

- Test tube or flask containing the gas sample to be tested
- Limewater
- Delivery tube or dropper/pipette to bubble the gas

Procedure

1. Carefully introduce the gas sample into a test tube containing a small amount of **clear limewater**.
2. Shake the test tube or allow the gas to bubble through the limewater for a short time.

Observation and Conclusion

Observation:

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Conclusion:

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C. Testing Procedure for Oxygen Gas

Principle

Oxygen is a colorless and odorless gas that is not flammable itself, but it is necessary for combustion (burning). The characteristic test uses a glowing splint, which reignites brightly when exposed to a high concentration of oxygen.

Safety Precautions

- Handle the lighted splint with care.

Apparatus and Materials

- Test tube containing the gas sample to be tested
- Wooden splint
- Matches or a lighter

Procedure

1. Light the wooden splint and allow it to burn for a moment, then blow out the flame so that the tip is still glowing red.
2. Carefully insert the glowing splint into the test tube containing the gas.

Observation and Conclusion

Observation:

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Conclusion:

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